
NEXTXR DIGITAL TWIN

Data Model

Relational schema, the Neo4j graph, and the seam between them

Generated 7 September 2026, from the running system

Repository Tejesh3305/Goalcert_Digital-Twin

Source docs/ER-DIAGRAM.md

Status Reflects the deployed implementation, not a design proposal

Generated from the live schema on 2026-09-07. Every table, column and row count below was introspected from the running databases rather than transcribed from a design document, so this reflects what the system actually stores.

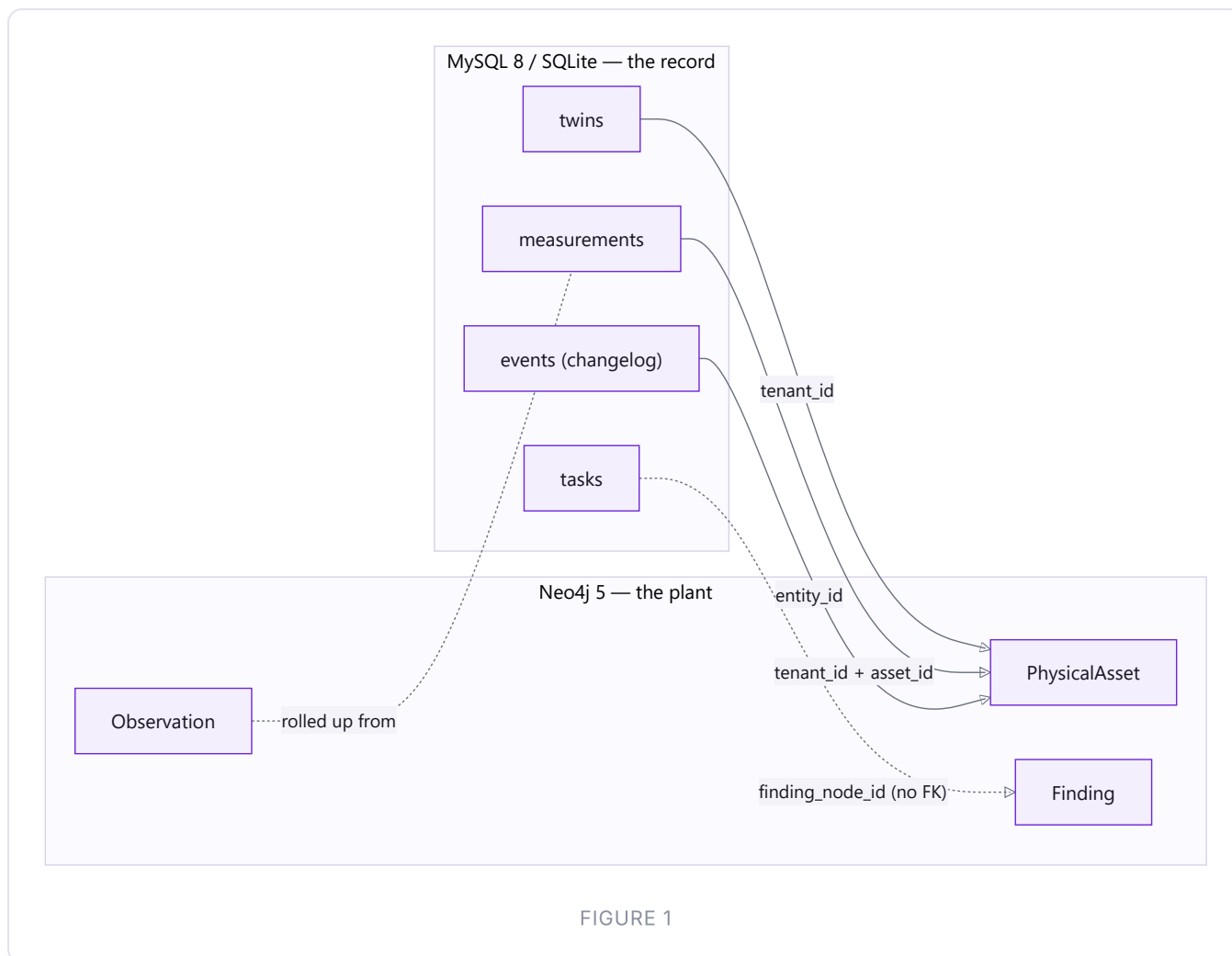
Source of truth for the relational schema is `nextxr-ontology/db/schema.py` ; the graph schema is provisioned by `nextxr-ontology/graph/schema.py` .

1. The polyglot split

This is the first thing to understand, because it is the design decision every other one follows from. The platform uses **two databases with two different jobs**, and they are deliberately not joined at the database layer.

	Holds	Answers
Neo4j (graph)	entities, relationships, findings	<i>What is the plant, and what is wrong with it</i>
MySQL / SQLite (relational)	identity, dispatch, telemetry history, registry	<i>Who did what about it, and when</i>

The seam between them is the pair `(tenant_id, node_id)` . A `tasks` row points at a Finding by `finding_node_id` and **copies nothing** from it. Dropping every relational row would lose the dispatch history and not one byte of twin state; dropping the graph would lose the plant model and leave the audit trail intact.



There are no declared foreign keys anywhere in the relational schema. This was verified, not assumed: `PRAGMA foreign_key_list` returns zero rows for every table. Referential integrity is enforced in the service layer (`work/service.py` , `identity/service.py`) and supported by 13 indexes on the work store alone. The relationships drawn below are therefore **logical**, and a diagram that showed them as enforced constraints would be lying about the guarantees. This is a deliberate trade — the schema targets both SQLite and MySQL through one DDL path — but it does mean orphan rows are possible if a write bypasses the service layer.

2. Identity and tenancy

The account model, shown as two diagrams rather than one: the four tables that define *who exists* answer different questions from the six that record *how they proved it*, and a single ten-entity diagram is wide enough that its column names stop being readable in print.

2a — Accounts and tenancy. `organizations` owns everything, and `org_tenants` is what binds a twin to an organisation — the join every authorisation check walks.

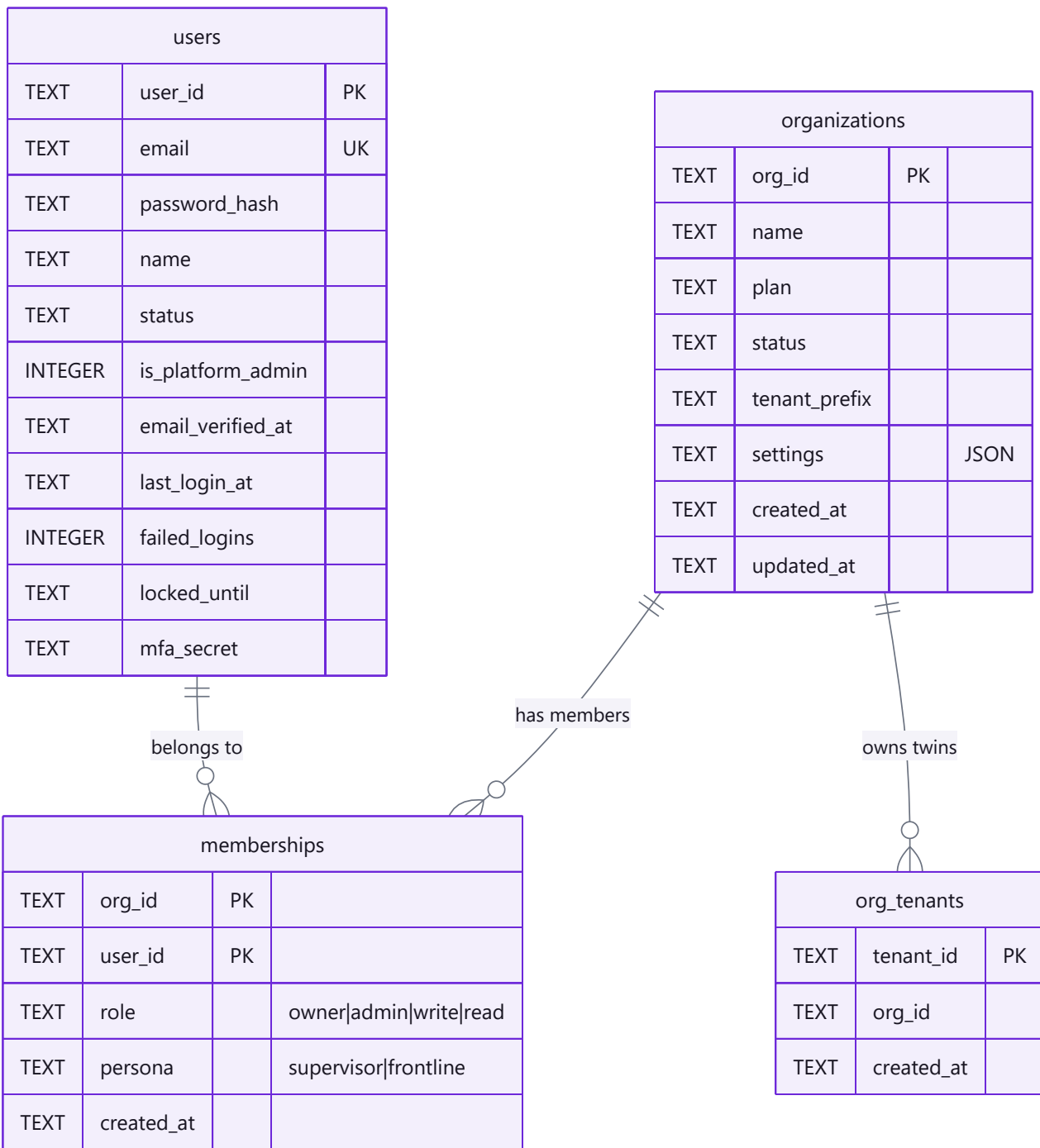


FIGURE 2

2b — Credentials, sessions and audit. Everything here is a *hash* or a *record of use*: no plaintext credential is stored in any of these tables.

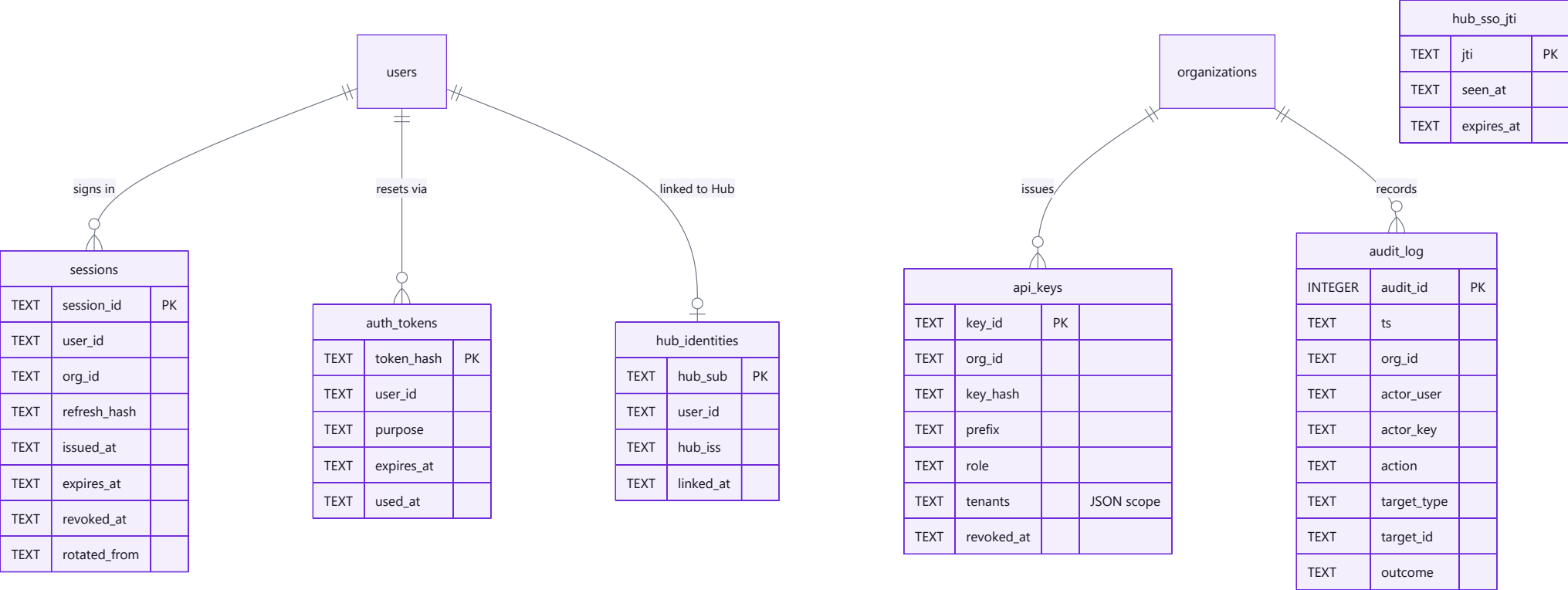


FIGURE 3 · SHOWN LANDSCAPE

`hub_sso_jti` stands alone by design — it is a replay guard, recording each Goalcert Hub SSO ticket id once so the same ticket cannot be presented twice.

The two role vocabularies

`memberships` carries **two orthogonal columns**, and collapsing them is the change this design exists to prevent:

column	vocabulary	answers	governs
<code>role</code>	owner / admin / write / read	what data may you touch	tenancy, via <code>server/tenancy.py</code>
<code>persona</code>	supervisor / frontline	what job do you do	dispatch, via <code>work/authority.py</code>

A supervisor who may only read the twin is an ordinary account — they dispatch people, they do not edit the plant model. One column cannot express both.

Known gap. Over HTTP the two are not fully independent: `server/auth.py` refuses every `POST` from a read-only principal before any persona check runs, so a `role='read'` supervisor cannot assign. `work/seed.py` issues `write`, so shipped accounts work. Flagged here because the schema promises a separation the middleware does not yet honour.

3. Work and dispatch

The fault-to-fix loop. This is the half of the product that records **what people did** about what the twin detected.

teams			
TEXT	team_id	PK	
TEXT	org_id		
TEXT	name		
TEXT	site		
TEXT	shift		
TEXT	supervisor_id		-> users
TEXT	created_at		

work_rules		
TEXT	rule_id	PK
TEXT	org_id	
TEXT	tenant_id	
TEXT	behavior_glob	
TEXT	min_severity	
TEXT	priority	
TEXT	scenario_id	
INTEGER	enabled	
INTEGER	seq	

tasks			
TEXT	task_id	PK	
TEXT	org_id		
TEXT	tenant_id		-> the twin
TEXT	code		TSK-0001
TEXT	title		
TEXT	detail		
TEXT	finding_node_id		-> Neo4j Finding
TEXT	asset_node_id		-> Neo4j asset
TEXT	asset_name		
TEXT	behavior_id		
TEXT	severity		
TEXT	scenario_id		-> procedure
TEXT	status		open assigned in_progress blocked resolved closed
TEXT	priority		
TEXT	assignee_id		-> users
TEXT	assigned_by		-> users
TEXT	team_id		
TEXT	source		twin_finding manual
REAL	score		
INTEGER	xp_awarded		
TEXT	resolution		
TEXT	changelog_ref		
TEXT	created_at		
TEXT	assigned_at		
TEXT	started_at		
TEXT	completed_at		

team_members		
TEXT	team_id	PK
TEXT	user_id	PK
TEXT	org_id	
TEXT	created_at	

scenario_runs			
TEXT	run_id	PK	
TEXT	org_id		
TEXT	user_id		
TEXT	task_id		

task_events			
TEXT	task_id	PK	
TEXT	event_id	PK	
TEXT	org_id		
TEXT	created_at		

xp_ledger	
TEXT	task_id
TEXT	asset_id
TEXT	finding_id
TEXT	behavior_id
TEXT	scenario_id
TEXT	created_at
TEXT	updated_at

staffed by

assigned within

decides which findings raise

transitions

awards

graded by

TEXT	event_id	PK	
TEXT	task_id		
TEXT	org_id		
TEXT	kind		
TEXT	actor_id		
TEXT	actor_name		
TEXT	summary		
TEXT	payload		JSON
TEXT	created_at		

TEXT	entry_id	PK	
TEXT	org_id		
TEXT	user_id		
TEXT	task_id		
TEXT	reason		
INTEGER	points		
TEXT	detail		
TEXT	created_at		

TEXT	scenario_id		
TEXT	mode		guided practice
TEXT	status		
INTEGER	step		
INTEGER	total_steps		
REAL	score		
INTEGER	passed		
INTEGER	hints_used		
INTEGER	wrong_steps		
TEXT	transcript		JSON
TEXT	started_at		
TEXT	completed_at		

FIGURE 4

Three design points the shape encodes

task_events is **append-only**. A **status** column remembers only the present. "Who assigned this, when, and what happened next" is the question asked after an incident, so every transition appends a row in the same call that changes the status.

xp_ledger is a **ledger, not a counter**. A total that can only be incremented cannot be explained, audited or corrected. The total is a **SUM**, and **(user_id, task_id, reason)** is unique so a double award is refused by the database rather than by a check-then-write race in Python.

Scoring is server-side. **scenario_runs.score** is written by **scenario/guided.py** from the answer key, which never leaves the server. If the page held the answers, the score would be a claim the browser made about itself.

4. Twin registry, content and telemetry

4a — Registry and published content. What a twin *is*, and the artefacts generated from it.

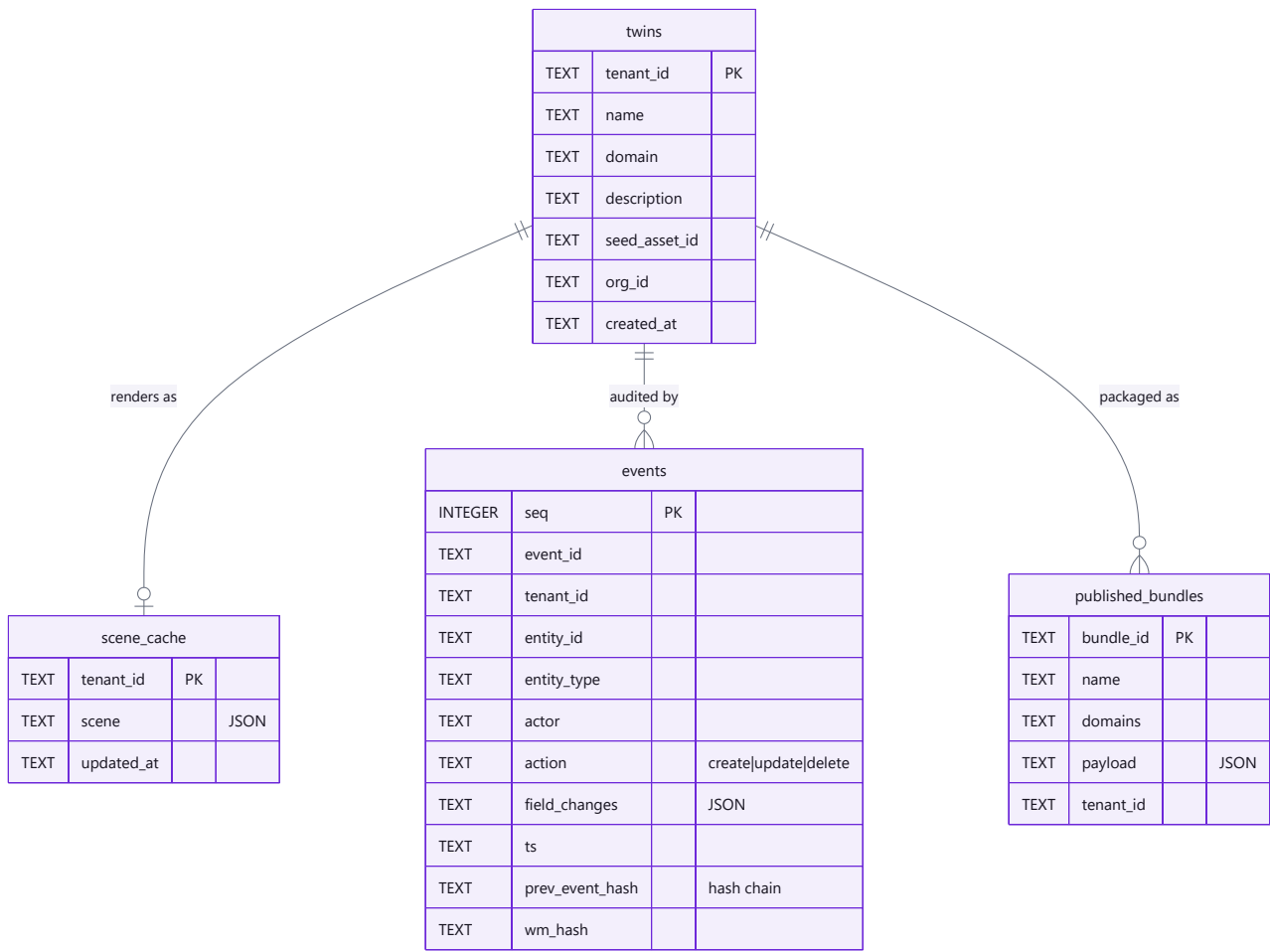


FIGURE 5

4b — Ingest, telemetry and jobs. How readings arrive, who is allowed to send them, and the asynchronous work they trigger.

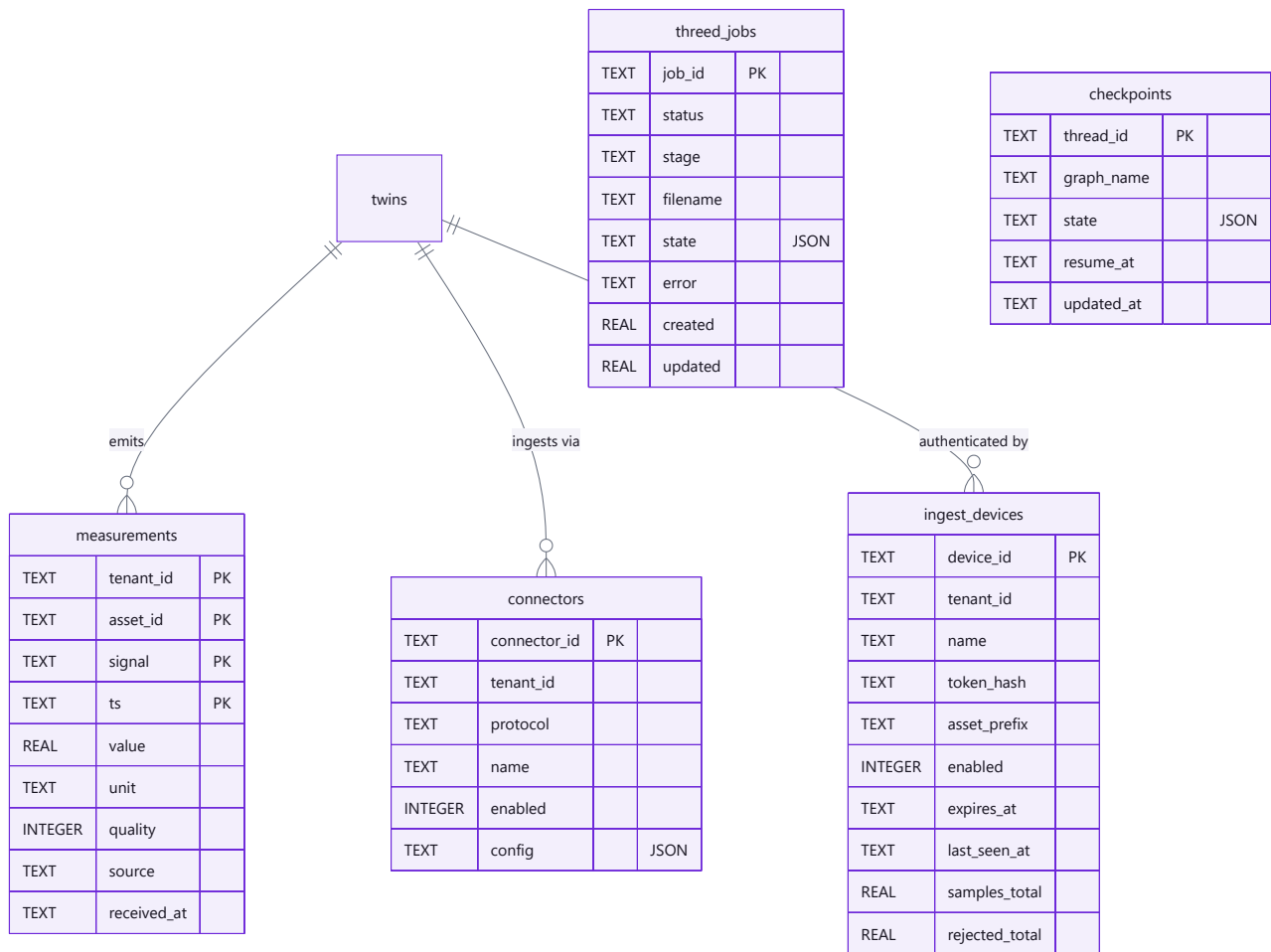


FIGURE 6

threed_jobs and **checkpoints** carry no **tenant_id**: a reconstruction job and an agent checkpoint are keyed by their own opaque id and reached only by the caller that created them.

measurements has a four-column primary key — (**tenant_id**, **asset_id**, **signal**, **ts**) . That is what makes ingest idempotent: replaying a batch overwrites rather than duplicating.

events is hash-chained. Each row carries **prev_event_hash** , so the change log is tamper-evident rather than merely append-only.

5. The graph model (Neo4j 5)

Eleven node categories, every one uniquely keyed by (**tenantId**, **id**) — that composite is the multi-tenancy boundary, enforced by a uniqueness constraint per label. Verified live: **11 constraints, 25 indexes**.

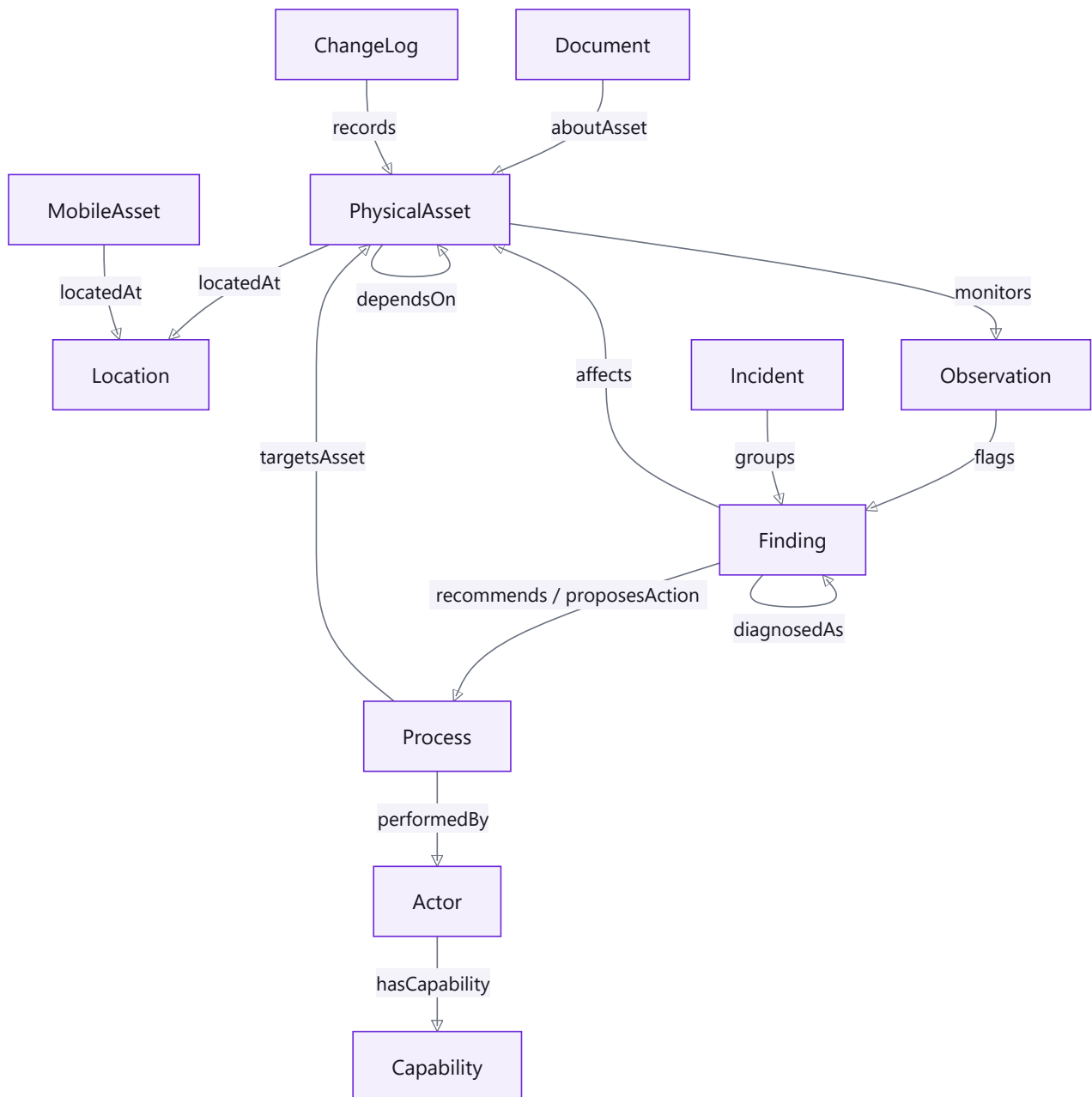


FIGURE 7

Node categories: Actor, Capability, ChangeLog, Document, Finding, Incident, Location, MobileAsset, Observation, PhysicalAsset, Process.

The relationship vocabulary is not ad-hoc — it comes from the ontology (`platform/nxr-properties.ttl`), which declares 25+ `owl:ObjectProperty` terms including `locatedAt`, `containedIn`, `hasPart`, `partOf`, `hasPort`, `connectsTo`, `dependsOn`, `feeds`, `monitors`, `observedAs`, `flags`, `diagnosedAs`, `indicatesFailureMode`, `recommends`, `proposesAction`, `assignedTo`, `performedBy`, `targetsAsset`. Domain packs (`packs/hvac`, `packs/edm`, ...) extend the class hierarchy but reuse this predicate set.

The Finding is the hinge of the whole product. A behaviour evaluates telemetry, writes a `Finding` into the graph, and `work/from_finding.py` turns qualifying Findings into `tasks` rows. That is the only place

the two databases meet, and it is one-directional: closing a task writes the Finding back to resolved, but no graph node ever holds a `task_id`.

6. Physical deployment of the stores

Logical store	Local dev	Production
identity, work, twins, changelog, bundles, checkpoints, scenes, threed, devices, connectors, historian	one SQLite file each, under <code>nextxr-ontology/data/</code>	one RDS MySQL 8 database , all stores sharing a transaction domain
graph	Neo4j in Docker (<code>neo4j_data</code> volume)	Neo4j 5 Community on EC2 , EBS-backed
blobs (GLBs, 3-D artifacts)	local filesystem	S3
event bus	Redis in Docker, in-memory fallback	ElastiCache Redis 7

`db/core.py` picks the backend from `NXR_DATABASE_URL`: set it and every store is MySQL; unset and each store is its own SQLite file. The DDL is written SQLite-flavoured and translated for MySQL, which is why `datetime('now')` and `INSERT OR IGNORE` do not appear anywhere in `store.py` files.

Migrations are ledgered in `schema_migrations` and applied by `python -m db.migrations`. Current head: **0006_work_graph**.

7. Live row counts (2026-09-07)

Useful as a sanity baseline; these are development volumes, not production.

Table	Rows
historian.measurements	34,347
changelog.events	13,961
identity.sessions	100
identity.audit_log	190
checkpoints.checkpoints	57
twins.twins	16
threed.threed_jobs	11
work.task_events	12
work.scenario_runs	8
identity.org_tenants	8
scenes.scene_cache	5
work.tasks	3
identity.users / memberships	3 / 3
identity.organizations	2
work.xp_ledger	2
work.teams / team_members / work_rules	1 / 1 / 1
devices.ingest_devices	1
bundles.published_bundles	3
connectors.connectors / identity.api_keys	0 / 0

The graph is currently near-empty — 3 nodes, 0 relationships, against 16 registered twins. The relational side persisted across restarts; the Neo4j volume did not retain content. Constraints and indexes are provisioned, so this is a data-loading gap rather than a schema one, but it means no Finding currently exists and the three `tasks` rows are seed fixtures with empty `tenant_id` and `finding_node_id`. Rebuild via `twins/seed.py::seed_demo_twins()`.